📘 How to Compile and Run `exoTest.c` (GCC + OpenMP)

This document explains how to compile and run a C program named `exoTest.c` using GCC (MinGW-w64) on Windows.

📄 1. Prerequisites

Before starting, make sure you have:

- GCC installed (MinGW-w64)
- `gcc` available in PATH
- A C file named:

```
exoTest.c
```

⚙️ 2. Compile the program

Open PowerShell or CMD in the folder where your file is located:

```
cd C:\Users\sidoa\Downloads\CHP
```

Then compile using:

```
gcc -fopenmp exoTest.c -o exoTest.exe
```

📌 Explanation

- `gcc` → GNU Compiler
- `-fopenmp` → enables OpenMP (for parallel programming)
- `exoTest.c` → source file
- `-o exoTest.exe` → output executable file

⚠️ Important: Never use the same name for input and output.

▶ 3. Run the program

After successful compilation:

```
.\exoTest.exe
```

🧠 4. Full Workflow (One Line)

You can compile and run directly:

```
gcc -fopenmp exoTest.c -o exoTest.exe && .\exoTest.exe
```

❌ Common Errors

1. Same input and output file

```
gcc: fatal error: input file is the same as output file
```

✓ Fix:

```
gcc exoTest.c -o exoTest.exe
```

---

2. gcc not recognized

✓ Fix:

- Add MinGW `bin` folder to PATH

Example:

```
C:\mingw64\bin
```

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3. OpenMP not working

✓ Ensure you use:

```
-fopenmp
```

🎯 Summary

Step	Command
Compile	`gcc -fopenmp exoTest.c -o exoTest.exe`
Run	`./exoTest.exe`

🚀 Done

Now you can compile and run any C O